Development of a novel high-pressure triaxial apparatus and investigation of the undrained behavior of sand in ultra-deep sea environment

Shao-Heng He, Zhen-Yu Yin*, Zhi Ding, Rui-Dong Li

Shao-Heng He, PhD, Professor, Department of Civil Engineering, Hangzhou City University, No. 48 Huzhou street, Hangzhou 310015, China, E-mail address: heshaoheng@zju.edu.cn

Zhen-Yu Yin, PhD, Professor, Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, No. 11 Yuk Choi Road, Hung Hom, Kowloon, Hong Kong 999077, China, E-mail address: zhenyu.yin@polyu.edu.hk (corresponding author)

Zhi Ding, PhD, Professor, Department of Civil Engineering, Hangzhou City University, No. 48 Huzhou street, Hangzhou 310015, China, E-mail address: dingz@hzcu.edu.cn

Rui-Dong Li, PhD candidate, Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, No. 11 Yuk Choi Road, Hung Hom, Kowloon, Hong Kong 999077, China, E-mail address: 22072096r@connect.polyu.hk

Abstract: Deep-sea mining is typically conducted on the seabed with water-depths exceeding several thousand meters, necessitating accurate understanding of soil mechanical behavior under ultra-high back pressure (b_p) while preserving conventional effective stress states. However, current high-pressure triaxial systems fail to achieve sufficient measurement accuracy for effective stress, and investigations on undrained behavior of sand under ultra-high back pressure remain scarce. To overcome this fundamental constraint, we propose a testing method that incorporates an auxiliary high-precision differential pressure sensor into high-pressure devices, enabling direct measurement of effective stress and substantially improving measurement accuracy. Moreover, the principles of calibration procedures and additional techniques to further enhance accuracy are demonstrated. To isolate the role of b_p, undrained monotonic and cyclic tests are performed on quartz sand under b_p ranging from 300kPa to 29MPa utilizing this enhanced high-pressure triaxial apparatus. The results demonstrate that b_p variations neither alter the slope of the critical state line in p'-q space nor significantly affect the evolution of effective stress ratios during shear, confirming the validity of the effective stress-based mechanical principle under ultra-high hydrostatic pressure. Furthermore, an energy-based analytical framework reveals a unique relationship between normalized energy dissipation and pore pressure evolution during cyclic loading regardless of b_p.

Keywords: High-pressure triaxial; sand; back pressure; shear behavior

1. Introduction

A substantial proportion of the earth's hydrocarbon and mineral resources are located in deep-sea environments (Cordes et al., 2016, Zhang et al., 2019). Technological advancements now enable the extraction of oil and natural gas from reservoirs at water depths exceeding 3,000 meters (Hu et al., 2013), as exemplified by ultra-deep water operations in the Gulf of Mexico (Cordes et al., 2016). Similarly, deep-sea mining has gained strategic importance due to vast mineral deposits found at depths of 1–6.5 kilometers (Koschinsky et al., 2018, Leng et al., 2021), with routine operations now being conducted on the seabed at a depth of around 3,000 meters underwater. During mining operations, mechanical disturbances on seabed from mining vehicles or hydraulic extraction processes can generate excess pore pressure, triggering instability and potential seabed failure (Leng et al., 2021). In ultra-deep sea environment, the hydrostatic pressure is extremely high. The physical and mechanical properties (e.g., density, bulk modulus, boiling point, and viscosity coefficient) of water under ultra-high pressure undergo significant changes. These changes present unique challenges in predictions for geotechnical engineering. For instance, at a water depth of 3 kilometers in the deep sea, the hydrostatic pressure can reach 30 MPa, the volume of water under this condition will be compressed by approximately 1.5~2% compared to that under atmospheric pressure. Previous studies have reported that pore fluid properties can significantly influence the compressibility, shear strength, and excess pore pressure generation capacity of soil specimens (Ratnaweera and Meegoda, 2006, Cabalar and Hasan, 2013). However, studies on the mechanical behavior of sand under ultra-high back pressure remain very limited. Consequently, investigating the complex effects of changes in the physical properties of water induced by ultra-high hydrostatic pressure on soil mechanical behavior is of paramount importance for the design of deep-sea engineering.

The operation of mining vehicles on the seabed surface (Koschinsky et al., 2018) necessitates a thorough understanding of soil undrained behavior under combined conditions of normal effective stresses (e.g., $p'\approx 100$ kPa) and ultra-high back pressures (up to 30 MPa) for successful deep-sea mining. Numerous studies have examined the effects of back pressure on soil mechanics (Brand, 1975, Allam and Sridharan, 1980, Miyazaki et al., 2011, Hyodo et al., 2013), demonstrating its significant influence on excess pore pressure generation during undrained shear, thereby altering effective stress and undrained strength in granular materials. Experimental research, particularly undrained stress-controlled cyclic triaxial tests with different back pressures (Xia and Hu, 1991, Naghavi and El Naggar, 2019) shows that sand liquefaction resistance increases with higher back pressures. However, these studies were limited to back pressures below 1 MPa, offering insights for shallow-sea engineering but limited applicability to deep-sea conditions.

Conducting experiments under ultra-high back pressures while maintaining normal effective stresses presents greater challenges, primarily due to equipment constraints. A key technical barrier lies in achieving precise effective stress control under ultra-high back pressures, as conventional methods relying on measurements of larger-range confining and back pressure sensors face inherent limitations. Specifically, large-range pressure sensors exhibit notably low measurement accuracy, typically 0.1%–0.15% of the full-scale range, which is insufficient for accurately maintaining effective stress within normal ranges. For specimens with low effective stress, measurement errors from large-range sensors can cause significant deviations in test results from the actual conditions. Recently, Sun et al. (2024) and Sun et al. (2023) conducted undrained monotonic triaxial tests on silt sand and cohesive soil with back pressures ranging from 100 kPa to 29 MPa using a commercial high-pressure triaxial apparatus. Their results highlighted the substantial role of back pressure in excess pore pressure evolution and undrained strength, though no effect on critical state stress ratio was observed. Critically, their study also did not resolve the challenge of ensuring adequate measurement accuracy for normal effective stresses under ultra-high back pressures, casting doubt on the reliability of the findings. The inherent precision constraints of current high-pressure triaxial devices hinder experimental studies on deep-sea soils under conventional effective stresses, and the mechanism governing ultra-high back pressure effects remains insufficiently clarified.

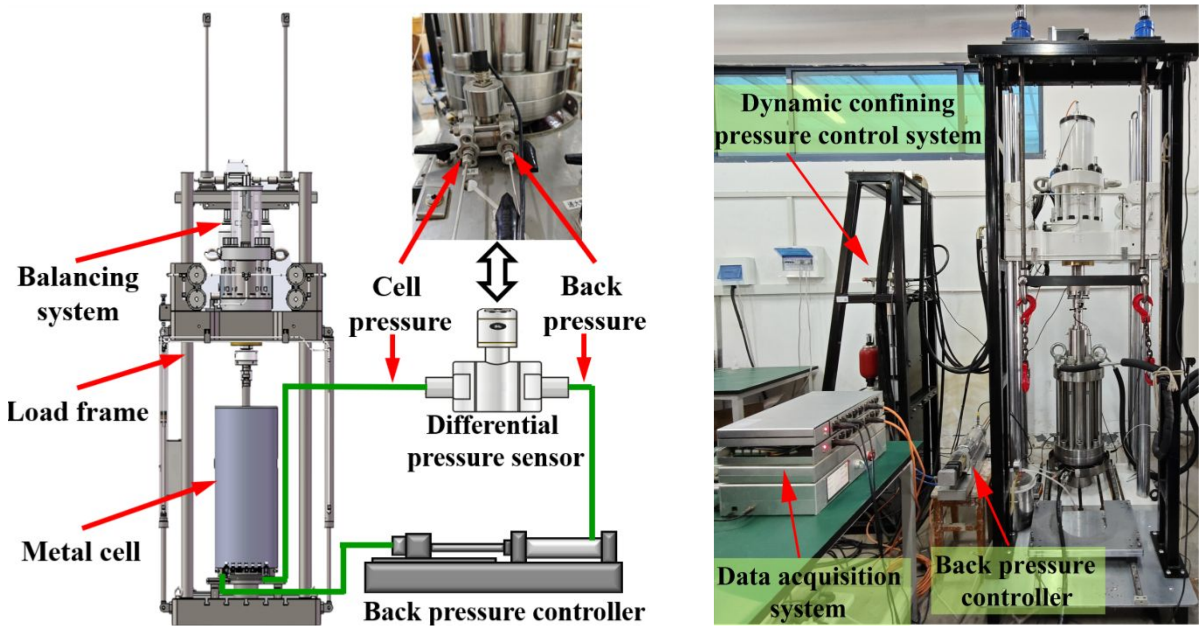
To address the above-mentioned inherent limitations, we proposed an innovative testing methodology that substantially improves effective stress measurement accuracy in high-pressure triaxial set-ups. This study provided a comprehensive demonstration of the methodological principle, calibration protocol, and supplementary techniques designed to improve testing precision for low-effective-stress specimens subjected to ultra-high back pressures. The proposed methodology was experimentally validated through a series of undrained monotonic and cyclic loading tests on Fujian sand. These tests systematically revealed the fundamental influence of ultra-high back pressure on the mechanical behavior of Fujian sand. Furthermore, an energy-based analytical framework was employed to investigate liquefaction responses under cyclic loading conditions, yielding insights for liquefaction prediction.

2. Novel high-pressure triaxial testing idea and structure design

2.1 Design idea of the novel high-pressure triaxial testing method

Most conventional studies are limited to the mechanical behavior of shallow-sea sand, largely owing to limitations in experimental facilities. The development of high-precision high-pressure triaxial apparatus and the exploration of mechanical properties of deep-sea sand under high back pressure are of great significance for advancing the fundamental theories of soil mechanics.

Technical specifications from leading global pressure volume controller (PVC) manufacturers, such as GDS company in the UK, indicate that the best measurement accuracy of a pressure sensor in a PVC is approximately 0.1-0.15% of its full scale range. The use of confining and back pressure sensors in traditional high-pressure triaxial devices, which have a range of 32 MPa, allows for only a best accuracy of 64 kPa in measuring effective stress (i.e., the difference between confining and back pressures). This measurement accuracy represents the accumulated maximum error across the sensor's full measurement range during prolonged testing. However, pressure sensors of PVC can typically achieve measurement resolutions < 0.4 kPa, even under conditions with a wide measurement range. This measurement resolution refers to the smallest change in pressure that the sensor can detect and distinguish (see Fig. 2). Although with high measurement resolution, the low measurement accuracy of PVC sensors in high-pressure setup renders such systems inadequate for precise effective stress control. Consequently, traditional high-pressure triaxial systems are better suited for investigating the mechanical properties of deep-earth soils with high effective stress (typically larger 1MPa) rather than shallow deep-sea soils. Addressing this technological shortfall requires developing novel measurement protocols, which are a prerequisite for advancing our understanding of deep-sea soil behavior.



(a) (b)

Fig. 1. Schematic for the main-body of high-pressure test set-up: (a) Schematic for the placement of the differential pressure sensor; (b) Main-body of the set-up

The low accuracy of large-range pressure sensors in high-pressure set-ups is an inherent and unavoidable limitation. The conventional effective stress measurement method in high-pressure set-ups, relying on low-precision confining and back pressure sensors, fails to precisely measure mean effective stress, significantly compromising the reliability of test results within the normal effective stress range. To address this critical limitation, this study incorporates an auxiliary differential pressure sensor with a high-accuracy and small-range (400 kPa range, 0.4 kPa accuracy) into the triaxial system. As illustrated in Fig. 1(a), the sensor is hydraulically connected to both the triaxial metallic cell chamber and back pressure volume controller (PVC), enabling direct measurement and control of the net pressure difference (i.e., confining pressure - back pressure) that fundamentally represents effective stress.

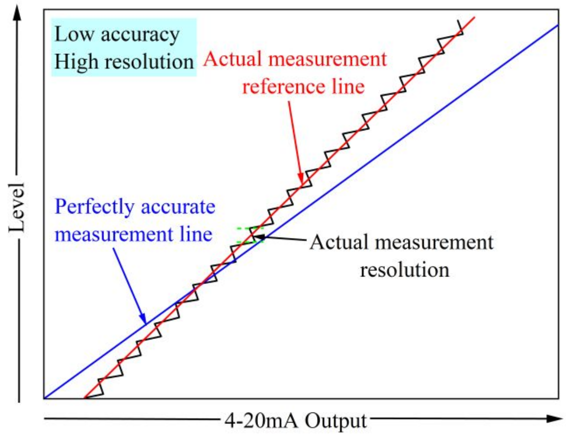


Fig. 2. Schematic for the typical measurement result of a pressure sensor with low accuracy and high resolution conditions

This study developed an innovative testing scheme that synergizes the high accuracy of differential pressure sensors with the fine resolution of PVC pressure sensors to accurately measure effective stress while maintaining constant total confining pressure (σ_3) during undrained shear. The protocol involves a meticulously controlled saturation and consolidation phase, where confining pressure and effective stress are incrementally increased at a predetermined rate until target values are attained. Real-time multi-sensor feedback dynamically adjusts back pressure to maintain synchronization throughout this process. After consolidation, precise effective stress is achieved by the feedback of the differential pressure sensor, and the finial confining pressure sensor reading is recorded as a baseline reference. In the subsequent undrained shear stage, this confining pressure reference value was held constant by the feedback of the confining pressure sensor, ensuring the total stress (σ_3) on the specimen remains stable during loading. The high resolution of the confining pressure sensor (e.g., < 0.4kPa) ensures precise control at the constant reference value, as any fluctuation in confining pressure exceeding 0.4 kPa can be detected and corrected. Since there is no increase in confining pressure after consolidation, the further increment in measurement error is nearly zero, as non-linear errors do not change abruptly and only become significant with large pressure increments (see Fig. 2). Meanwhile, the differential pressure sensor was used solely to accurately measure excess pore pressure during undrained shear stage. This framework facilitates a total effective stress measurement accuracy of 0.4kPa and allows for an accurate quantification of the mechanical response characteristics exhibited by sediments under deep-sea conditions. A movable overload protection diaphragm is installed inside the differential pressure sensor. Once the pressure difference exceeds the upper limit of the measuring range, the diaphragm shifts sideways, bringing the external isolation diaphragm on the high-pressure side into close contact with the inner wall of the chamber. This blocks further pressure transmission to the sensitive component and prevents sensor damage caused by overpressure. In addition, the system will automatically limit the current when the sensor output current exceeds the safety threshold, so as to avoid damage resulting from overcurrent.

2.2 Additional improvement strategies in structural design and calibration

The apparatus shown in Fig. 1(a) has been preliminarily used to investigate the temperature-dependent behavior of sand under ultra-high back pressure with accurate effective stress control (He et al., 2025), while other specific innovative instrument designs for high-precision measurements will be detailed in this study. Fig. 1(b) further illustrates the overall structure of the high-pressure triaxial apparatus. Compared to conventional high-pressure triaxial apparatuses, the current apparatus has been enhanced in several key aspects to better facilitate the precise measurement of the mechanical properties of ultra-deep-sea soils. Under conditions of high hydrostatic pressure, maintaining a constant confining pressure within the metal cell during the cyclic loading process is particularly challenging. To address this issue, this study utilizes a dynamic confining PVC capable of applying cyclic confining pressure at a frequency of 20 Hz, as shown in Figs. 3(a-b). Compared to conventional confining PVC, this dynamic system can rapidly stabilize the confining pressure during loading, even under high pressure and cyclic loading.

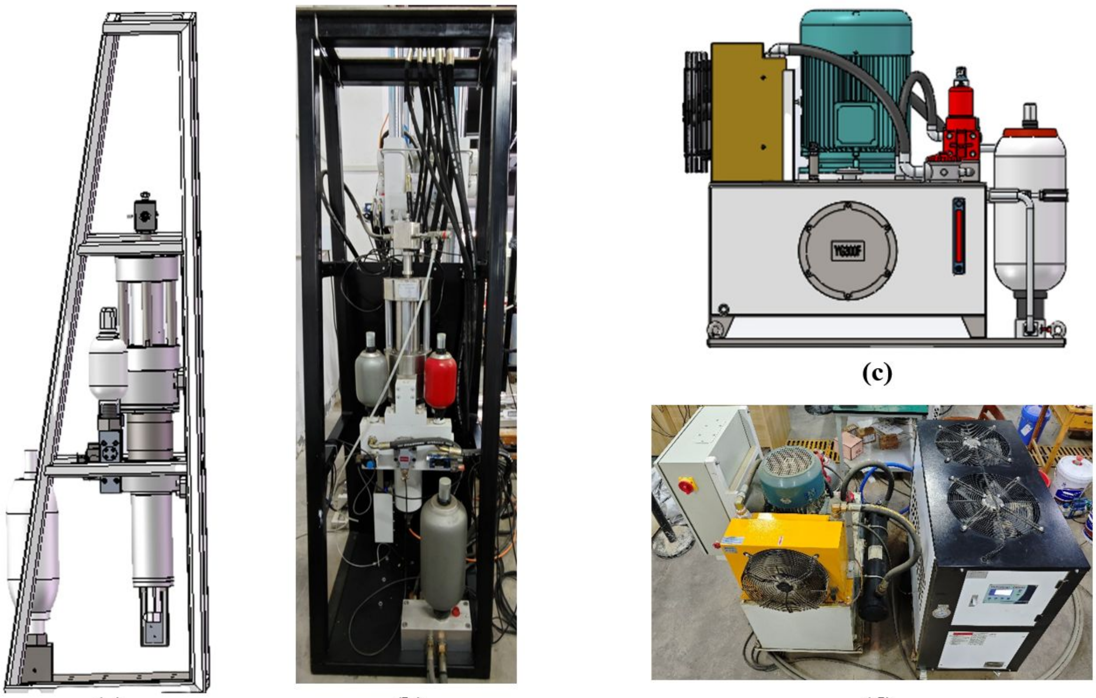


Fig. 3. Schematic for additional parts of high-pressure set-up: (a) and (b) Schematic for the dynamic PVC; (c) and (d) Schematic for the hydraulic station

Both axial loading and the confining PVC are powered by a high-capacity hydraulic station, as depicted in Figs. 3(c-d). To ensure stable control of axial forces, a balance system is integrated into the loading frame (see Fig. 1(a)). The fluid within the middle chamber of the balancing system is connected to the pressure chamber via the loading rod. As the loading rod moves in a piston-like manner, the fluid in the balancing system flows correspondingly within the chamber surrounding the piston, maintaining a constant total fluid volume. This stabilizes both the confining pressure and axial force. Additionally, fluctuations in the confining pressure around the specimen in the pressure chamber, caused by axial piston motion, can be further mitigated through pressure compensation using a dynamic confining pressure control system with high-speed regulation capability. Under high confining pressure conditions, the axial loading rod encounters significant resistance from the ambient cell pressure when penetrating the pressure chamber. Another critical function of the balancing system, therefore, is to assist the axial loading rod in resisting the isotropic confining pressure within the chamber, ensuring that the forces acting on both the axial loading rod and frame correspond to the deviatoric stress. Additionally, a high-precision 5 kN load cell, designed for underwater operation, is placed directly within the confining pressure chamber in contact with the specimen during loading.

In conventional high-pressure triaxial systems, steel pipes (see Fig. 4(a)) are typically used for the pipeline system connected to the specimen due to the extreme pressure requirements (≥30 MPa). This pipeline system includes hydraulic connections to the back pressure volume controller and an exhaust passage to the atmosphere for creating a vacuum between the suction rubber cap and the specimen. The pipeline connected to the suction rubber cap must withstand a confining pressure of 32 MPa during loading, necessitating that the material of the pipeline possesses exceptionally high strength to avoid being crushed and blocked under high pressure. However, the excessive rigidity of the steel pipes can exert a considerable supporting force on the specimen during loading, leading to significant discrepancies between the test results and the actual mechanical properties of the material. Therefore, both traditional plastic and steel pipes used in both conventional and high-pressure triaxial devices are no longer suitable. After testing various materials, this study identified a polyether ether ketone (PEEK) material that does not exhibit significant compressive deformation or blockage under 32 MPa high pressure and is sufficiently flexible like plastic (see Fig.4(b)) to avoid applying additional supporting force to the specimen during loading. The PEEK pipes connected to the suction cap have a direct diameter of 1.6 mm, an inner diameter of 0.1 mm, and a wall thickness of 0.75 mm. The pipes connected to the back pressure have a thinner wall thickness since the internal back pressure provides support within the pipeline, as shown in Fig. 4(c). Preliminary leakage tests were conducted at a pressure of 32 MPa before formal triaxial tests to confirm no internal water leakage and ensure satisfactory sealing performance.

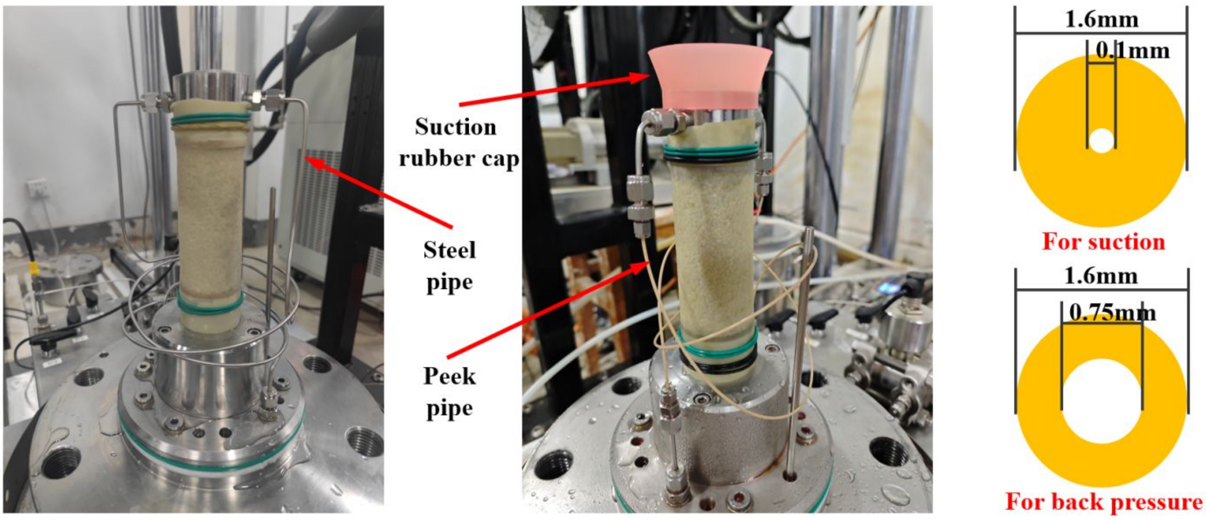


Fig. 4. Schematic for the piping system: (a) Specimen with steel pipe; (b) Specimen with PEEK pipe; (c) Sizes for the PEEK pipe

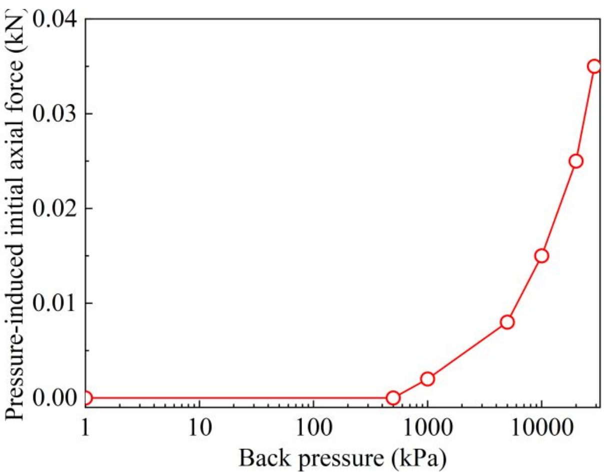


Fig. 5. Pressure-induced initial axial forced versus b_p

Under ultra-high hydrostatic pressure testing conditions, the strain gauges in the load cell experience an initial additional deformation due to the high hydrostatic pressure, leading to an initial force reading unrelated to the specimen. To calibrate for this initial additional increment of force in the sensor caused by ultra-high hydrostatic pressure at different back pressures, this study examined the changes in the force sensor readings under various back pressures without specimen installation. The initial force reading induced by different hydrostatic pressures are as shown in Fig. 5(a). It is clearly observed that as b_p increases, especially when b_p exceeds 1MPa, the initial axial force reading due to the hydrostatic pressure considerably rises. Based on the experimental results presented in Fig. 5(a), this study recalibrated the sensor readings by subtracting the initial additional force caused by the hydrostatic pressure in the experiments conducted at different back pressures.

The values of back pressure and confining pressure can be directly obtained from the pressure sensors in the pressure volume control systems, while the pore water pressure can be measured using the pore pressure sensor installed at the base of the triaxial apparatus. The axial strain (ε_a) is calculated as $\varepsilon_a=\Delta h/h$, where Δh is the axial displacement measured by a linear grating encoder installed in the axial actuator, and h is the height of the specimen. The axial load (f_a) is measured by the underwater load cell. Accordingly, the axial stress (σ_a) is calculated as $\sigma_a= f_a/A$, where A is the current area of the specimen. Under undrained shear conditions, the area A is calculated by $A=V/(h-\Delta h)$ under undrained shear conditions, assuming a constant specimen volume V.

3. Test applications

Extensive field investigations have documented the prevalence of sandy sediments in ultra-deep sea environments beyond 3,000 meters water depth (Smith et al., 2023). Consequently, this study focuses on sandy soil as the test subject to exemplify the application of enhanced high-pressure triaxial testing, aiming to investigate its mechanical properties in ultra-deep sea environments.

3.1 Preparation of sand specimen and test method

Fujian sand, recognized as the standard sand of China, was used as the testin. The air pluviation method was employed to prepare triaxial specimens with a diameter of 50 mm and a height of 100 mm. De-aired water and carbon dioxide were passed through the specimen (He et al., 2026), followed by the application of back pressure saturation to achieve a very high saturation level, with the B-value exceeding 0.99, thereby eliminating the effects of saturation degree. After saturation, the specimen underwent isotropic consolidation to reach the desired mean effective stress.

Table 1. Physical properties of Fujian sand

Parameter	**Fujian Sand**
Specific gravity	2.65
Maximum void ratio	0.884
Minimum void ratio	0.512
Sphericity	0.896
Aspect ratio	0.720
Convexity	0.949
Roundness	0.625

Considering that the seabed surface sand sediment in deep-sea mining operations is typically in a relatively loose state with low effective stress, this study focuses on testing under a single condition of relatively loose relative density. After consolidation, specimens with a relative density corresponding to approximately 55% were used for shear tests, with the specific void ratios of each specimen recorded in Tables 2 and 3. To highlight the essential differences in the mechanical behavior of Fujian sand in deep-sea environments compared to terrestrial environments, this study conducted monotonic undrained tests under both conventional and ultra-high back pressures. Specifically, four back pressure values were selected: 300 kPa, 1 MPa, 15 MPa, and 29 MPa (see Table 2), with a shear rate of 0.1 mm/min. Strain-controlled cyclic triaxial tests are widely used to evaluate the liquefaction potential of soils, as studies by Silver and Park (1976) and Sitharam et al. (2012) strongly suggest that cyclic shear strain, rather than cyclic shear stress, governs both densification and liquefaction in sands. In this study, undrained strain-controlled cyclic tests were performed on specimens to assess their liquefaction resistance, with shear strain amplitudes (γ_{ampl}) of 0.07%, 0.1%, 0.2%, and 0.3% at a frequency of 0.1 Hz. Although the high-pressure triaxial apparatus adopted in this study is capable of applying cyclic axial loads at high frequencies, we still set the cyclic loading frequency to 0.1 Hz to ensure stable data collection and facilitate comparative analysis with previous research findings. Cyclic tests were conducted under back pressures of 300 kPa and 29 MPa, with specific details provided in Table 3. The monotonic and cyclic test results under b_p=300 kPa and 29 MPa were adopted from the study by He et al. (2025) for comparative analysis and re-examination of energy dissipation mechanisms across different b_p conditions.

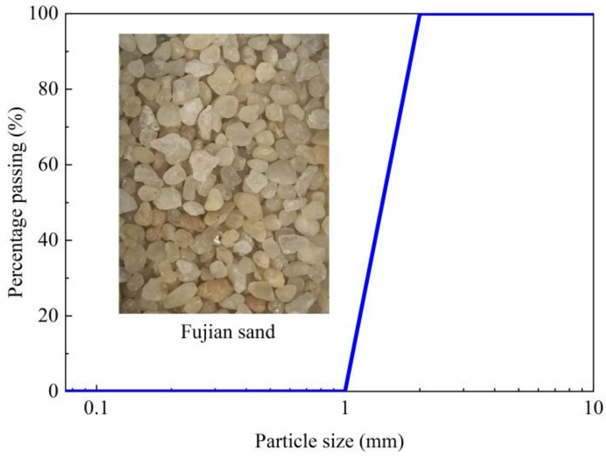


Fig. 6. Particle size distribution of Fujian sand

Table 2. Scheme of undrained monotonic tests

p'_0 (kPa)	b_p (kPa)	e	p'_0 (kPa)	b_p (kPa)	e
50	300	0.688	100	300	0.676
50	1000	0.687	100	1000	0.676
50	15000	0.689	100	15000	0.677
50	29000	0.690	100	29000	0.678
200	300	0.663			
200	1000	0.664			
200	15000	0.664			
200	29000	0.665			

Note: The symbol e denotes the void ratio of specimens after consolidation.

Table 3. Scheme of undrained cyclic tests

p'_0 (kPa)	γ_{ampl} (%)	b_p (kPa)	e	p'_0 (kPa)	γ_{ampl} (%)	b_p (kPa)	e
100	0.07	300	0.677	100	0.2	300	0.677
100	0.07	29000	0.679	100	0.2	29000	0.678
100	0.1	300	0.676	100	0.3	300	0.676
100	0.1	29000	0.678	100	0.3	29000	0.679

Note: The symbol e denotes the void ratio of specimens after consolidation.

3.3 b_p-dependent monotonic behavior of Fujian sand

Fig. 7 illustrates the variations of deviatoric stress and excess pore pressure as a function of axial strain under different back pressures (b_p). The results demonstrate that regardless of initial effective stress p'_0, increasing b_p leads to greater accumulation of excess pore pressure during shear, consequently resulting in reduced deviatoric stress. When b_p ranges between 300 kPa and 1 MPa, its variation exhibits an insignificant influence on both the stress-strain relationship and the evolution of excess pore pressure. However, as b_p rises to 15 MPa and particularly reaches 29 MPa, a higher b_p significantly enhances shear-induced positive pore pressure generation while markedly decreasing deviatoric stress (q). Sieve analysis conducted on the tested specimens demonstrates that back pressure exerts negligible influence on particle breakage, even at 29 MPa. A lower undrained strength observed in sand under ultra-high back pressure indicates that conventional triaxial tests may be unsafe for deep-sea engineering design, highlighting the necessity of conducting triaxial tests under actual back pressure conditions.

Fig. 8 illustrates the evolution of the effective stress ratio (q/p') with axial strain during shear under different back pressures. As axial strain increases, q/p' initially rises rapidly to a peak value, followed by a slight decrease with further increases in axial strain. This post-peak reduction becomes progressively more pronounced with higher values of p'_0. Notably, the q/p' curves under varying b_p values largely overlap regardless of p'_0, with only slight variations observed during the initial loading phase (i.e., axial strain range: 0-5%). During this initial loading phase, specimens subjected to a higher b_p, particularly at b_p=29MPa, exhibit marginally lower q/p' values compared to those under lower b_p. The insignificant effect of b_p on the q/p' evolutions indicates that b_p has a minimal impact on the development of effective friction angle in shear. This suggests that variations in b_p do not influence the fundamental mechanisms of inter-particle interactions, such as friction, interlocking, and biting. The results in Fig. 8 confirm the validity of the effective stress-based mechanical principle under ultra-high hydrostatic pressure.

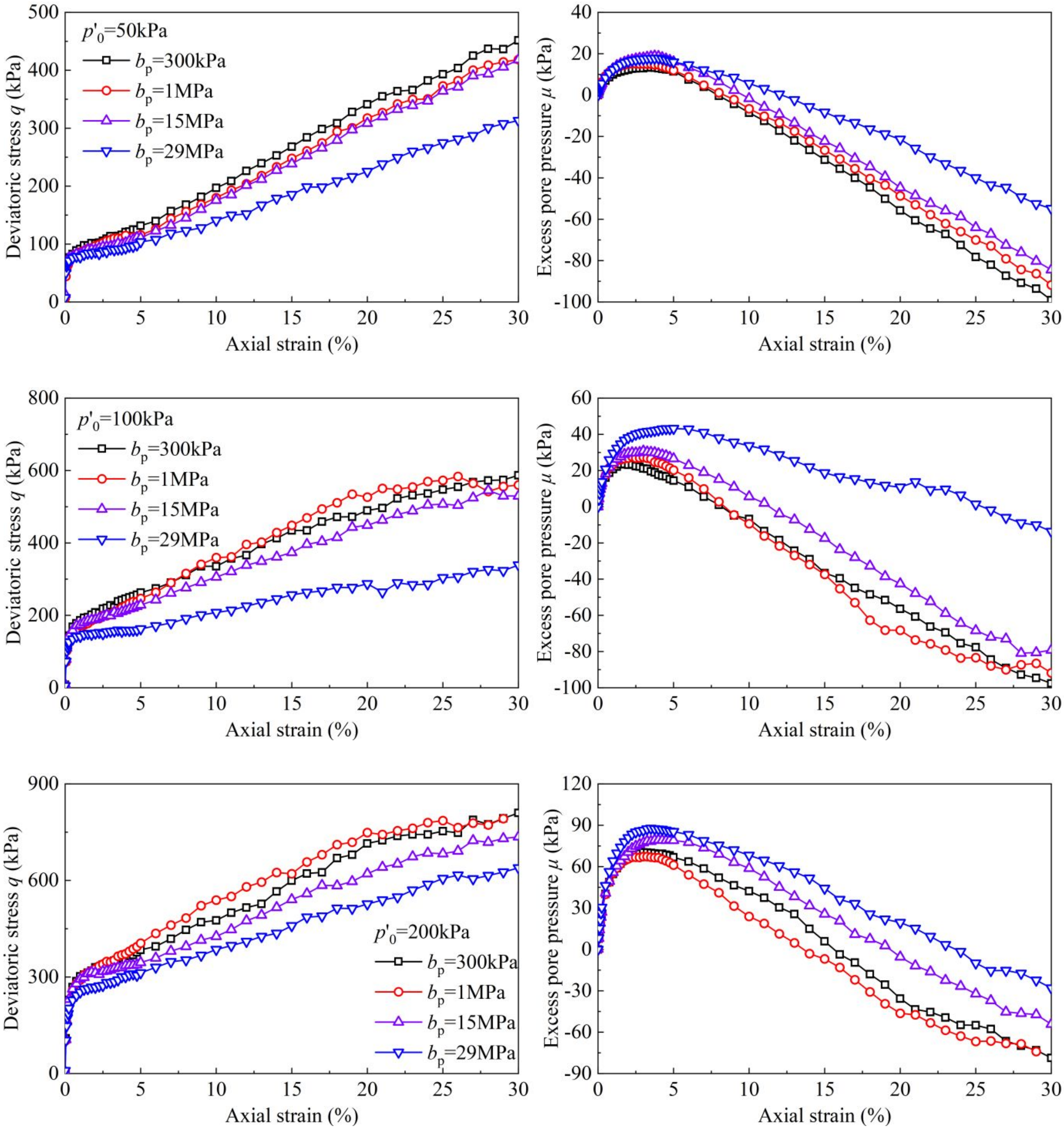
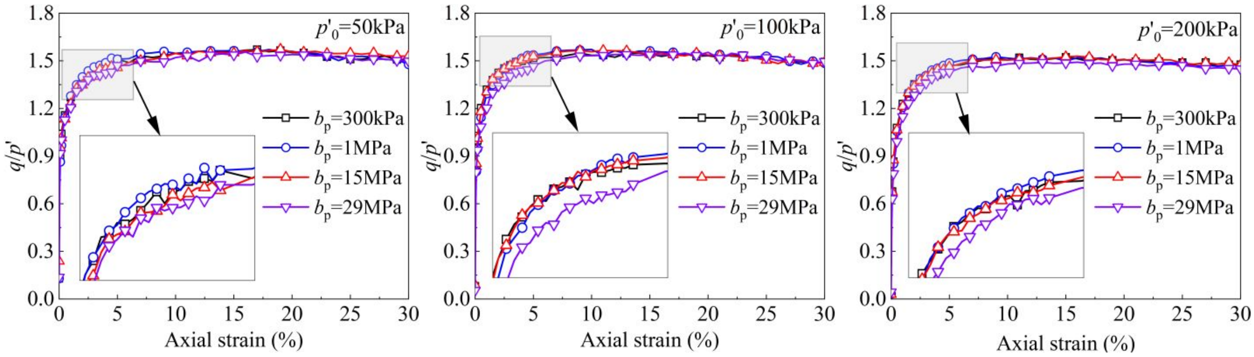


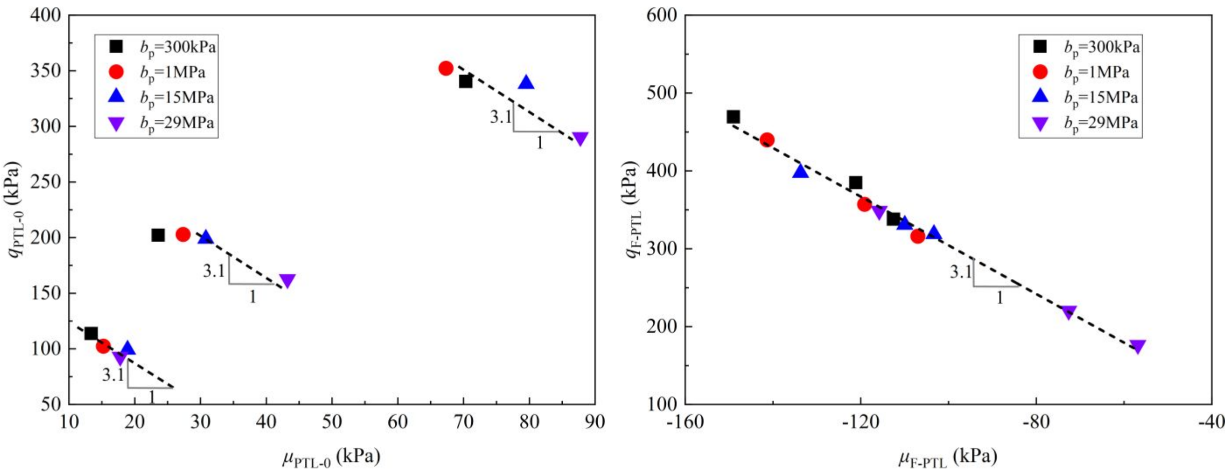
Fig. 7. Stress-strain relation and excess pore pressure evolution under different b_p



(a) (b) (c)

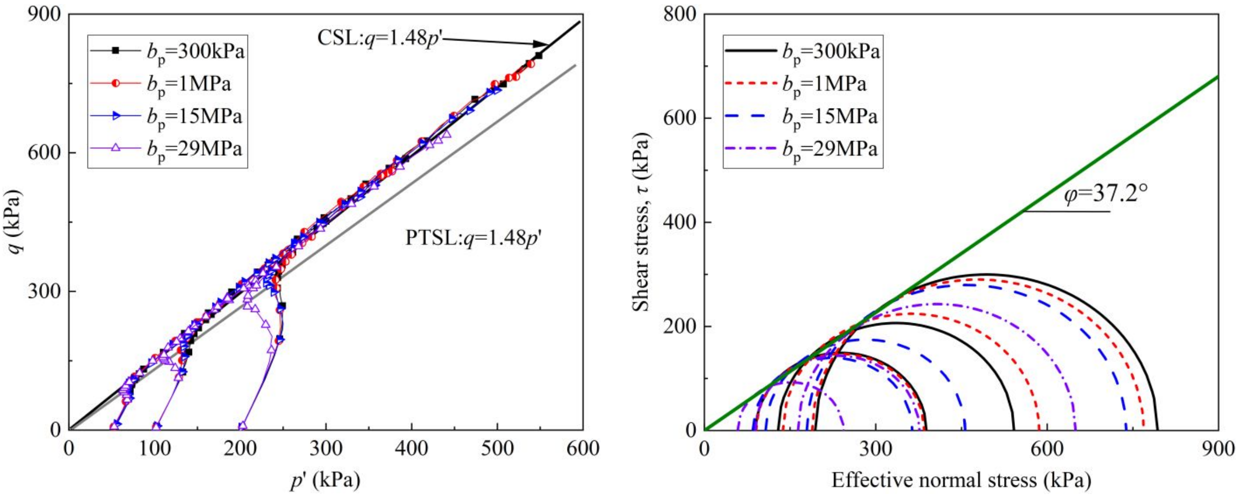
Fig. 8. Effective stress ratio evolution under different b_p: (a) p'_0=50kPa; (b) p'_0=100kPa; (c) p'_0=200kPa

The differences in excess pore pressure and deviatoric stress during the phase transition state compared to the initial state are denoted as $\mu_{PTL\text{-}0}$ and $q_{PTL\text{-}0}$, respectively. Similarly, the differences in excess pore pressure and deviatoric stress in the final state compared to the phase transition state are denoted as $\mu_{F\text{-}PTL}$ and $q_{F\text{-}PTL}$, respectively. The $\mu_{PTL\text{-}0}$-$q_{PTL\text{-}0}$ and $\mu_{F\text{-}PTL}$-$q_{F\text{-}PTL}$ curves are utilized to investigate the impact of changes in shear-induced u, caused by variations in b_p during the compressive and dilative phases, on shear strength. In Fig. 9(a), the relationship between $\mu_{PTL\text{-}0}$-$q_{PTL\text{-}0}$ is influenced by the initial mean effective stress p'_0. For specimens with larger p'_0, a higher initial strength and greater generation of positive u are observed before reaching the phase transition state. However, during the dilative phase, the increase in deviatoric stress due to the decrease in u at a higher b_p is independent of p'_0, with all data across different p'_0 values aligning on a single curve. As shown in Fig. 9(b), the change in deviatoric stress exhibits a linear relation with respect to the variation in excess pore pressure, and the slope remains consistent across both the compressive and dilative stages. This indicates that the change in undrained strength of sandy soil at different b_p values is directly related to the shear-induced u, which depends on b_p. The variation in excess pore pressure due to changes in b_p alters the effective stress of the specimens, resulting in a proportional change in strength.



(a) (b)

Fig. 9. Degradation in q caused by the b_p-dependent shear-induced u: (a) Compressive stage; (b) Dilative stage

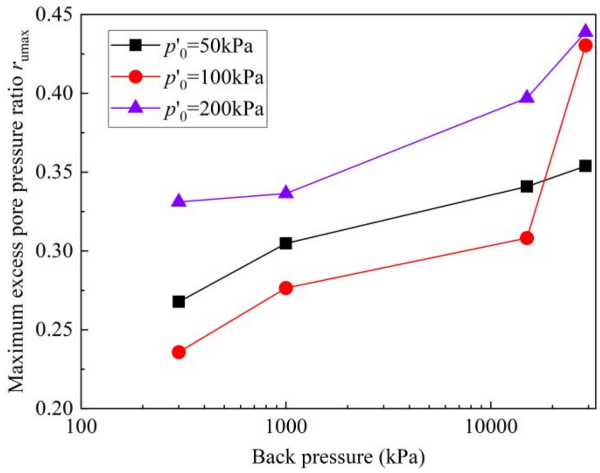


(a) (b)

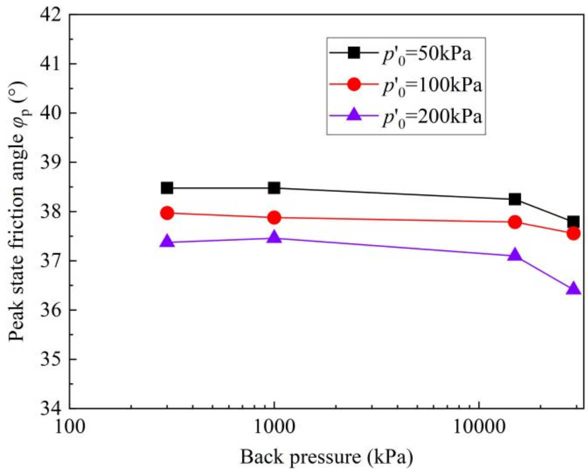
Fig. 10. Stress paths and strength envelopes; (a) Critical state line in p'-q plane; (b) Mohr–Coulomb failure envelope line

Fig. 10 presents the effective stress paths of all experiments under varying back pressures (b_p). Notably, regardless of b_p values, the critical state lines for all specimens converge to a unified trajectory, confirming that b_p variations do not alter fundamental particle-scale friction mechanisms and the critical state stress ratio. Instead, b_p primarily influences the stress path evolution before reaching the critical state by changing the excess pore pressure (u) generation, thereby dictating the effective stress evolution. Specifically, a higher b_p enhances compressive trend prior to critical state, shifting stress paths toward lower deviatoric stress (q) and mean effective stress (p'). Furthermore, the phase transition state line (PTSL) also exhibits b_p-independent convergence. Sun et al. (2024) conducted undrained monotonic triaxial shear tests on silty sand under various back pressures and reached conclusions consistent with this study. Their results confirm that back pressure exerts a negligible influence on the critical state stress ratio of sandy soils. In contrast, the undrained triaxial tests on clay conducted by Sun et al. (2023) demonstrated that the critical state stress ratio of cohesive soil increases significantly when the back pressure rises to 25 MPa. This reveals that the influence of back pressure on the critical state behavior of soils is highly dependent on soil type and internal microstructural features, and thus merits further systematic exploration.

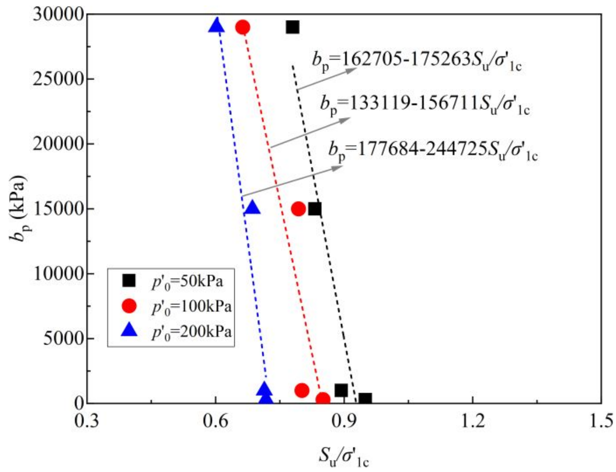
Mohr–Coulomb failure envelopes based on the maximum effective stress ratio $(q/p')_{max}$ failure criterion are presented in Fig. 10 (b) for undrained shear tests conducted under different back pressure (b_p) and p'_0 values. Although variations in back pressure lead to reduced effective normal and shear stresses at failure, all experimental data approximately converge to a unified Mohr-Coulomb failure envelope, with a friction angle of approximately 37.2°. Notably, the findings in Fig. 10 demonstrate that changes in b_p, even under deep-sea conditions equivalent to 3,000 meters of hydrostatic pressure, have an insignificant effect on the effective-stress-based mechanical parameters of sand, such as the effective stress ratios.



(a)



(b)



(c)

Fig. 11. b_p-dependent behaviors: (a) r_{umax} versus b_p; (b) φ_p versus b_p; (c) b_p versus S_u/σ'_{1c}

Fig. 11(a) illustrates the evolution of the maximum excess pore pressure ratio (r_{umax}=u_{max}/p'_0) during shear as a function of back pressure. The results indicate a significant increase in r_{umax} with higher back pressures, reflecting an enhanced capacity for excess pore pressure generation in the specimens. Fig. 11(b) further explores the correlation between the peak friction angle (φ_p) and parameters b_p and p'_0. Note that this φ_p is derived by $\sin\varphi_p = \frac{3\eta_p}{6+\eta_p}$ (η_p is the effective stress ratio at peak state). As p'_0 increases, the peak friction angle of the sand experiences a slight decrease. Notably, a further increase in b_p beyond 15 MPa also produces an effect similar to that of an increase in p'_0, resulting in a slight reduction in the peak friction angle. The undrained shear strength ratio (S_{us}/σ'_{1c}) is defined as the normalized value of undrained shear strength relative to the initial major effective principal stress at consolidation (σ'_{1c}), expressed by (Tsukamoto et al., 2009):

$$\frac{S_{us}}{\sigma'_{1c}} = \frac{q_{PTS}}{2}\cos\varphi_{PTS}\frac{1}{\sigma'_{1c}} \tag{1}$$

where q_{PTS} and φ_{PTS} represent the deviatoric stress and internal friction angle, respectively, at the phase transformation state. Parameter S_{us}/σ'_{1c} serves as a crucial indicator of shear strength in soil mechanics. As demonstrated in Fig. 13(c), S_{us}/σ'_{1c} exhibits a monotonic decreasing trend with increasing b_p, independent of p'_0. Although the influence of b□ on effective stress ratios at both peak and critical states is negligible, its direct reduction effect on the normalized shear strength ratio remains an important consideration in geotechnical design to ensure adequate safety

Previous undrained triaxial tests under high back pressure have confirmed that increasing back pressure can modify the interparticle contact mechanism of sand. For example, Xia and Hu (1991) conducted a series of undrained cyclic triaxial tests on sand specimens at different back pressures. They pointed that a higher hydrostatic pressure imposes extra confinement and restricts particle movement. Consequently, under identical cyclic axial loading, specimens exhibit distinct particle movements, thereby generating different incremental excess pore water pressure during each loading cycle and ultimately yielding varied cyclic effective stress paths. Although back pressure hardly changes the macroscopic effective stress measured on specimen surfaces, it can adjust the microscopic interparticle force state inside samples. According to the study by Bahadori and Vuthaluru (2009), the relationship between the bulk modulus (K_f) of water and back pressure is depicted in Fig. 12, showing that an increase in back pressure results in a noticeable rise in K_f. The increase in bulk modulus of water under ultra-high back pressure may enhance the ability of pore water to generate pressure under the same strain conditions. Moreover, relevant undrained shear tests on silty sand conducted by Sun et al. (2024) further support this mechanism. Their research indicates that the differences in undrained shear behavior of silty sand under various back pressures are primarily governed by variations in shear-induced cumulative excess pore water pressure, which ultimately dominates the effective stress state of soils.

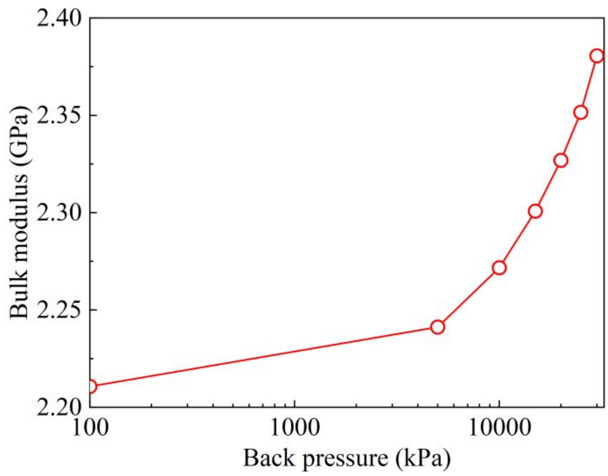


Fig. 12. Bulk modulus versus b_p (derived from Bahadori and Vuthaluru (2009))

3.5 b_p-dependent cyclic behavior of Fujian sand

The typical experimental results from a strain-controlled undrained cyclic loading test (representative case: γ_{ampl}=0.3%) are presented in Fig. 13. The excess pore pressure u gradually accumulates with the number of loading cycles N, ultimately reaching the initial mean effective stress (p'_0) and triggering liquefaction. This excess pore pressure development directly influences the deviatoric stress response shown in Fig. 13(b-c), where the cyclic amplitude of q exhibits rapid degradation due to effective stress reduction, eventually approaching zero at liquefaction onset. Compared to conventional back pressure conditions, specimens subjected to high back pressure exhibit a larger excess pore pressure accumulation rate with N, demonstrating a heightened capacity for excess pore pressure generation. This behavior aligns with observations from strain-controlled monotonic tests, where a larger u accumulation is observed under ultra-high b_p. Fig. 14 further illustrates u evolution across varying γ_{ampl}. Notably, ultra-high back pressure universally enhances u accumulation rates regardless of γ_{ampl}, with the effect becoming increasingly pronounced at smaller γ_{ampl}.

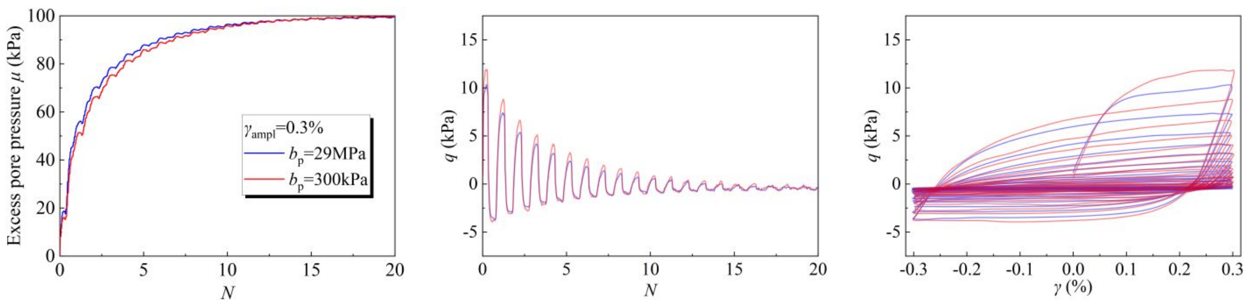
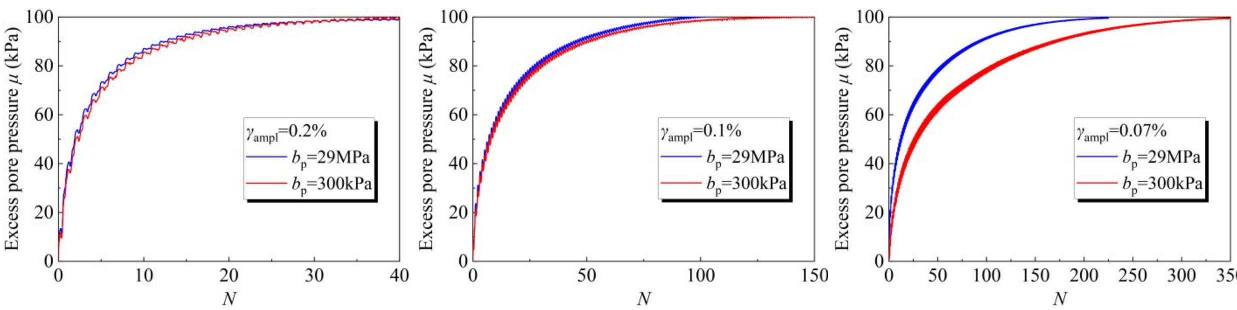


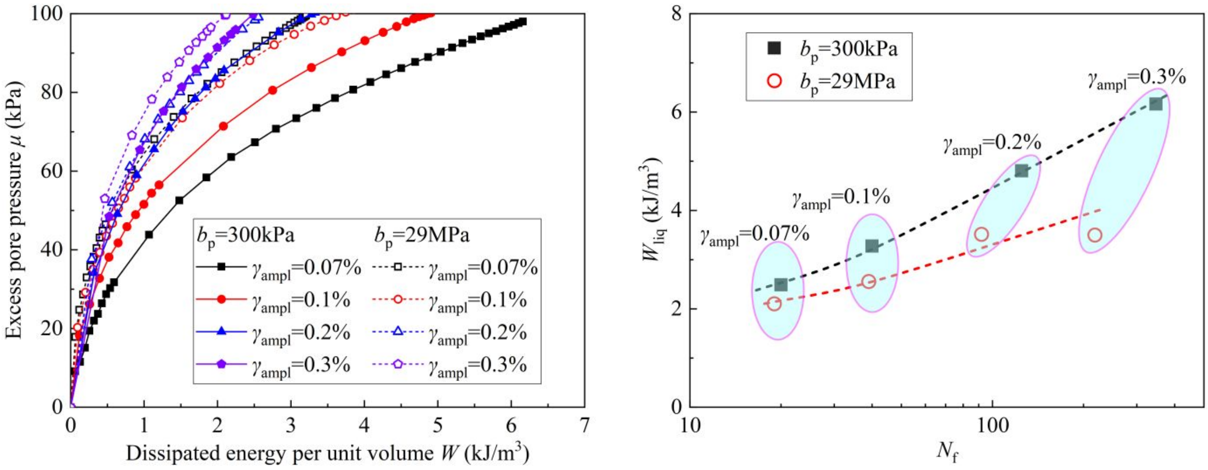
Fig. 13. Typical undrained test results during strain-controlled cyclic loading



(a) (b) (c)

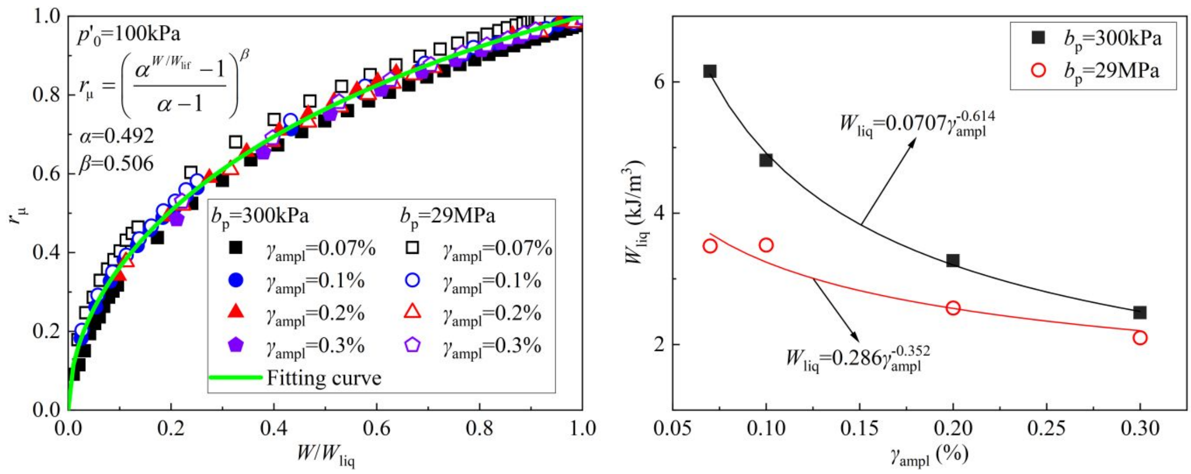
Fig. 14. u evolution under different b_p: (a) γ_{ampl}=0.2%; (b) γ_{ampl}=0.1%; (c) γ_{ampl}=0.07%

The energy-based methodology has become increasingly prevalent in soil liquefaction assessment due to its fundamental advantages over conventional stress- or strain-based approaches (Jafarian et al., 2012). As a scalar quantity, energy considers the entire spectrum of ground motion, incorporating both strain and stress, as well as material properties. The total energy accompanied by the rearrangement and movement of sand particles during liquefaction is considered to be a fixed amount under specific repetitive loading circumstances. The energy dissipated per unit volume (W) is defined as the area within the hysteresis loop developed during a cycle, which is directly related to the development of excess pore pressure (u) during the loading process. Fig. 15(a) shows the relationship between W and u for all experiments. It can be observed that the applied cyclic shear amplitude γ_{ampl} affects the relationship between the dissipated strain energy and the accumulation of excess pore pressure. As γ_{ampl} increases, a smaller W is observed to correspond to the same excess pore pressure. Notably, b_{p} also has a significant impact on the generation of pore water pressure and energy dissipation during cyclic loading. At a higher b_{p}, the specimen accumulates more excess pore pressure for the same amount of dissipated strain energy. This indicates that changes in b_{p} affect the energy dissipation mechanism of the sand specimens during cyclic loading, possibly due to b_{p}-induced changes in pore fluid density and compressibility. The increase in b_{p} enhances the bulk modulus of the pore water, thereby improving its ability to absorb energy and redistribute energy allocation during strain-controlled cycling. Consequently, during strain-controlled cyclic loading, more input energy is directed toward generating pore excess pressure, while the portion dissipated through strain energy (i.e., kinetic energy) decreases. Moreover, the altered accumulation rate of excess pore pressure during strain-controlled cyclic loading further alters the effective stress and the overall mechanical behavior of the specimens, leading to changes in their energy dissipation mechanisms.



(a) (b)

Fig. 15. Dissipated energy behaviors: (a) u versus W; (b) W_{liq} versus N_{f}



(a) (b)

Fig. 16. Normalized dissipated energy behaviors: (a) r_{μ} and W/W_{liq}; (b) W_{liq} versus γ_{ampl}

The number of loading cycles required to initiate liquefaction is denoted as N_{f}, whereas the cumulative dissipated strain energy necessary for liquefaction triggering is termed the energy capacity (W_{liq}). The correlations between W_{liq} and N_{f} under different values of γ_{ampl} are presented in Fig. 15(b). The data reveal an inverse relationship where increasing N_{f} corresponds to a lower γ_{ampl}, accompanied by a proportional rise in the energy capacity W_{liq}. The energy capacity W_{liq} of sand specimens is strongly correlated with liquefaction resistance, which is dominated by particle-scale contact mechanisms. The remarkable variation in W_{liq} caused by changes in b_{p} can be attributed to the evolution of interparticle contact behaviors. These alterations reshape the energy demand for particle motion, and further change the energy dissipation and transfer characteristics at the particle scale.

The strain energy ratio is designated for the ratio of W to the capacity energy (W_{liq}). The excess pore pressure ratio (r_{u}) is defined as the ratio of u to p'_0. As shown in Fig. 16(a), the r_{u}−W/W_{liq} relationship exhibits consistent trends across a wide range of γ_{ampl} and b_{p} values, with the r_{u}−W/W_{liq} data points located within a narrow band. This suggests a unique correlation between r_{u} and W/W_{liq} that is independent of variations in b_{p} and γ_{ampl}. To quantify this relationship, we employ the model proposed by Jafarian et al. (2012) (Eq. (4)), which demonstrates excellent agreement with the experimental data.

$$r_{\mathrm{u}}=\left(\frac{\alpha^{W/W_{\mathrm{liq}}}-1}{\alpha-1}\right)^{\beta} \tag{4}$$

where α and β are material constant. The findings in Fig. 16(a) indicate that the r_{u}−W/W_{liq} correlation could serve as a practical tool for predicting the evolution of excess pore pressure during seismic loading. Furthermore, Fig. 16(b) illustrates the energy dissipation mechanism governing the γ_{ampl}−W_{liq} relationship exhibits clear b_{p}-dependence. The results reveal that W_{liq} decreases exponentially with γ_{ampl}, while higher b_{p} values significantly reduce energy capacity.

4. Conclusions

This study proposes an innovative testing methodology that substantially improves effective stress measurement accuracy in high-pressure triaxial set-ups. Utilizing this enhanced high-pressure triaxial apparatus, undrained monotonic and cyclic tests were performed on Fujian sand to investigate the role of ultra-high back pressure on mechanical behaviors. The primary conclusions of this research are as follows:

(a) This study presents an innovative testing methodology that significantly enhances the control accuracy of effective stress in high-pressure triaxial devices. A key enhancement involves the integration of a high-accuracy and small-range differential pressure sensor directly linking the triaxial cell chamber and back pressure volume controller, enabling direct measurement and control of effective stress. By integrating the high accuracy of the differential pressure sensor with the high resolution of conventional large-range pressure sensors, this study proposes a hybrid measurement approach that enables precise control of effective stress, ensuring consistent maintenance of the target value throughout the entire loading test procedure.

(b) Ultra-high hydrostatic pressure induces an initial force reading in the load cell due to high-pressure-induced deformation of its internal strain gauge. This force reading, unrelated to the specimen, is systematically quantified through calibration tests under varying hydrostatic pressures (h_p). The results demonstrate that this spurious force reading increases with h_p, becoming significant when h_p>1MPa and necessitating pre-experimental calibration to isolate and subtract these high-pressure-induced effects prior to loading.

(c) With increasing back pressure (b_p) values, particularly when b_p exceeds 15 MPa, specimens generate greater shear-induced excess pore pressure under identical axial strain (ε_1), leading to reduced effective stress and consequently lower undrained shear strength. As b_p rises, the maximum excess pore pressure ratio during shearing increases, while the peak friction angle slightly decreases when b_p exceeds 15 MPa. Notably, in the q-p' plane, specimens under different b_p conditions all converge to a unique critical state line, though their stress paths are altered toward a more compressive side by b_p before reaching the critical state. Moreover, the failure surfaces under different b_p can be uniformly approximated by a Mohr-Coulomb strength envelope with a consistent internal friction angle. These findings demonstrate that b_p variation does not alter fundamental particle-scale friction mechanisms (e.g., intergranular friction and interlock) but changes the pore water pressure generation capacity during shearing, thereby solely diverging stress paths prior to reaching the critical state

(d) Under strain-controlled cyclic loading, specimens exposed to ultra-high back pressure (b_p) demonstrate markedly accelerated rate of excess pore pressure generation compared to conventional b_p conditions, resulting in earlier liquefaction onset. Furthermore, variations in b_p substantially alter the energy dissipation mechanism during cyclic loading. For identical strain energy dissipation, specimens under ultra-high b_p generate greater excess pore pressure. The energy-based framework can effectively characterize liquefaction responses across different b_p values, with all specimens showing a unique b_p-independent r_u−W/W_{liq} relation.

Acknowledgements

This research was financially supported by the Research Grants Council (RGC) of Hong Kong Special Administrative Region Government (HKSARG) of China (Grant no. 15220221, 15226822).

Competing interests statement

The authors declare there are no competing interests.

Data availability

Data generated or analyzed during this study are available from the corresponding author upon reasonable request.

References

Allam, M. & Sridharan, A. (1980) Influence of the Back Pressure Technique on the Shear Strength of Soils. *Geotechnical Testing Journal* **3(1)**:35-40.

Bahadori, A. & Vuthaluru, H. B. (2009) Prediction of bulk modulus and volumetric expansion coefficient of water for leak tightness test of pipelines. *International Journal of Pressure Vessels and Piping* **86(8)**:550-554.

Brand, E. W. (1975) Back Pressure Effects On The Undrained Strength Characteristics of Soft Clay. *Soils and Foundations* **15(2)**:1-16.

Cabalar, A. F. & Hasan, R. A. (2013) Compressional behaviour of various size/shape sand–clay mixtures with different pore fluids. *Engineering Geology* **164**:36-49.

Cordes, E. E., Jones, D. O. B., Schlacher, T. A., Amon, D. J., Bernardino, A. F., Brooke, S., Carney, R., Deleo, D. M., Dunlop, K. M., Escobar-Briones, E. G., Gates, A. R., Génio, L., Gobin, J., Henry, L.-A., Herrera, S., Hoyt, S., Joye, M., Kark, S., Mestre, N. C., Metaxas, A., Pfeifer, S., Sink, K., Sweetman, A. K. & Witte, U. (2016) Environmental Impacts of the Deep-Water Oil and Gas Industry: A Review to Guide Management Strategies. *Frontiers in Environmental Science* **4**.

He, S.-H., Yin, Z.-Y. & Ding, Z. (2025) Temperature-dependent undrained behaviour of sand in an ultra-deep marine environment. *Géotechnique*.

He, S.-H., Yin, Z.-Y., Ding, Z., Li, R.-D. & Liao, D. (2026) Drained deformation and non-coaxiality of granular materials under pure principal stress rotation: role of particle morphology. *Acta Geotechnica*.

Hu, W., Bao, J. & Hu, B. (2013) Trend and progress in global oil and gas exploration. *Petroleum Exploration and Development* **40(4)**:439-443.

Hyodo, M., Yoneda, J., Yoshimoto, N. & Nakata, Y. (2013) Mechanical and dissociation properties of methane hydrate-bearing sand in deep seabed. *Soils and Foundations* **53(2)**:299-314.

Jafarian, Y., Towhata, I., Baziar, M. H., Noorzad, A. & Bahmanpour, A. (2012) Strain energy based evaluation of liquefaction and residual pore water pressure in sands using cyclic torsional shear experiments. *Soil Dynamics and Earthquake Engineering* **35**:13-28.

Koschinsky, A., Heinrich, L., Boehnke, K., Cohrs, J. C., Markus, T., Shani, M., Singh, P., Smith Stegen, K. & Werner, W. (2018) Deep-sea mining: Interdisciplinary research on potential environmental, legal, economic, and societal implications. *Integrated Environmental Assessment and Management* **14(6)**:672-691.

Leng, D., Shao, S., Xie, Y., Wang, H. & Liu, G. (2021) A brief review of recent progress on deep sea mining vehicle. *Ocean Engineering* **228**:108565.

Miyazaki, K., Masui, A., Sakamoto, Y., Aoki, K., Tenma, N. & Yamaguchi, T. (2011) Triaxial compressive properties of artificial methane-hydrate-bearing sediment. *Journal of Geophysical Research: Solid Earth* **116(B6)**.

Naghavi, N. & El Naggar, M. H. (2019) Application of back pressure for saturation of soil samples in cyclic triaxial tests. In *Geotechnical Engineering in the XXI Century: Lessons learned and future challenges*.) IOS Press, pp. 115-123.

Ratnaweera, P. & Meegoda, J. (2006) Shear Strength and Stress-Strain behavior of Contaminated Soils. *Geotechnical Testing Journal* **29(2)**:133-140.

Silver, M. L. & Park, T. K. (1976) Liquefaction Potential Evaluated from Cyclic Strain-Controlled Properties Tests on Sands. *Soils and Foundations* **16(3)**:51-65.

Sitharam, T., Ravishankar, B. & Patil, S. (2012) Liquefaction and pore water pressure generation in sand: cyclic strain controlled triaxial tests. *International Journal of Geotechnical Earthquake Engineering*

(IJGEE) **3(1)**:57-85.

Sun, A., Ren, Y., Yang, G., Yu, L., Kong, G., Wang, N. & Yang, Q. (2023) Ultrahigh static pore pressure effects on the undrained shear behavior of deep-sea cohesive soil. *Applied Ocean Research* **130**:103404.

Sun, A., Yang, G., Ren, Y., Kong, G. & Yang, Q. (2024) Experimental study of ultra-high backpressure influence on the monotonic mechanical characteristic of silty sand. *Marine Georesources & Geotechnology* **42(5)**:496-505.

Tsukamoto, Y., Ishihara, K. & Harada, K. (2009) Evaluation of undrained shear strength of soils from field penetration tests. *Soils and Foundations* **49(1)**:11-23.

Xia, H. & Hu, T. (1991) Effects of Saturation and Back Pressure on Sand Liquefaction. *Journal of Geotechnical Engineering* **117(9)**:1347-1362.

Zhang, G., Qu, H., Chen, G., Zhao, C., Zhang, F., Yang, H., Zhao, Z. & Ma, M. (2019) Giant discoveries of oil and gas fields in global deepwaters in the past 40 years and the prospect of exploration. *Journal of Natural Gas Geoscience* **4(1)**:1-28.